VerifyChat: A Chat-Integrated Fact-Checking System for Developing Student Verification Mental Models

Sabbir Hossain Ujjal
Stevens Institute of Technology
sujjal@stevens.edu

Ayush Misra
Stevens Institute of Technology
amisra2@stevens.edu

Daniel Petrovsky
Stevens Institute of Technology
dpetrovs1@stevens.edu

Abstract

Students are increasingly relying on AI chatbots for academic tasks yet they lack systematic strategies for detecting factual errors or content known as "hallucinations" in AI-generated responses. Existing verification tools either automate fact-checking which requires leaving the chat interface entirely, creates friction in the process that suppresses adoption of verification. We present **VerifyChat**, a web-based chat system with an integrated claim-level verification panel that implements a *predict-then-reveal* interaction pattern: students commit to a judgment about which claims are inaccurate *before* automated verdicts are shown. We conducted a within-subjects study with undergraduate students ($N = 10$ recruited). Results show a huge improvement in hallucination-detection F1 (0.035 → 0.109), a sharp drop in NASA-TLX effort ratings (on average 3.315 → 1.91), and qualitative evidence of strategy shifts toward claim-level reasoning. A notable counterpoint—one participant showed F1 = 0 in both conditions while rating the tool 5/5 for usability—mirrors the "false security" effect documented for automated verification tools. We discuss implications for scaffolded AI literacy tools, the tension between usability and genuine engagement, and directions for longitudinal mental-model measurement. Demo Video of our system: VerifyChat Demo

Keywords

fact-checking, AI literacy, hallucination detection, cognitive forcing functions, metacognitive scaffolding, chatbots

1 Introduction

Large language model (LLM) chatbots such as ChatGPT are now deeply embedded in academic practice. Students rely on them for studying, writing assignments, and understanding new concepts. Yet the outputs of these systems are not always reliable: LLMs are known to *hallucinate*—producing confident, fluent, and plausible-sounding statements that are factually incorrect. Research shows that approximately 50% of ChatGPT solutions were mistakenly accepted as correct by students [10], and roughly 45% of students report copying AI-generated content directly without any verification [6]. Even students who are aware of hallucination risk continue to rely on AI without verification [17], indicating that awareness alone does not change behavior.

A second factor amplifies the first: *cognitive offloading*. When AI tools provide immediate, well-formed answers, users tend to outsource not only the task but also the evaluation of the information [8]. Fan et al. term this *metacognitive laziness*: students delegate self-monitoring and self-assessment to the AI, effectively offloading the judgment of whether the AI is correct [5]. Kosmyna et al. found that 83% of participants could not recall a single sentence from AI-generated text they had just read [9], suggesting that fluent output is processed so shallowly that critical evaluation rarely occurs. The OECD has flagged the erosion of students' self-regulatory skills as a direct consequence of uncritical AI use [13].

The HCI community has developed several verification tools to address this problem. HILL [12] assigns confidence scores to individual claims and surfaces external sources; RELIC [2] highlights self-consistency contradictions across multiple LLM outputs. Both were evaluated in CHI studies and demonstrated improved error identification—but neither has been released as a deployable artifact. Commercial tools such as Facticity AI [4] and Exa's [3] Hallucination Detector go further but require users to leave their chat interface, a friction barrier that suppresses adoption [8]. Crucially, all existing tools *verify for* students rather than *with* them: they automate the verdict without engaging students' own reasoning, which eventually risks producing the "false security" effect—positive usability ratings without genuine behavioral change [12]. No published system combines (a) automated claim-level verification with (b) metacognitive scaffolding (c) inside the natural chat workflow, and no study has measured whether such a combination changes students' verification *mental models* over time.

We present **VerifyChat**, a web-based chat system with an integrated verification panel that implements a *predict-then-reveal* interaction pattern. After each AI response, the system decomposes the response into atomic claims. Before automated verdicts are shown, students are prompted: *"Which claims do YOU think might be inaccurate?"* Only after students commit to a prediction does the system reveal verdicts alongside web-retrieved evidence snippets. This cognitive forcing function is grounded in Buçinca et al., who showed that requiring deliberation before seeing AI output significantly reduced overreliance in AI-assisted decision-making [1], and in Tankelevitch et al., who demonstrated that prompting self-assessment improves critical evaluation of AI outputs [16].

We conducted a within-subjects study ($N = 10$) in which participants completed a Roman-Empire history task in two conditions: (1) a standard chat baseline, and (2) the VerifyChat interface. Our research questions are:

- **RQ1**: Does VerifyChat help students more easily verify AI-generated responses during chatbot interactions?
- **RQ2**: How does using VerifyChat change students' mental models for evaluating the reliability of AI-generated content?
- **RQ3**: Does the predict-then-reveal scaffolding pattern improve students' ability to independently identify inaccurate claims?

Contributions

Our study makes main three interlocking contributions to HCI research:

(1) **Design**: The first chat-integrated, publicly deployable fact-checking system that embeds claim-level verification within the conversational workflow, eliminating context-switching friction.
(2) **Interaction Mechanism**: A predict-then-reveal cognitive forcing function applied to AI fact-checking—a novel transfer of Buçinca et al.'s paradigm to the domain of claim-level verification.
(3) **Empirical Measurement**: A behavioral metric which includes prediction accuracy, F1 score between student predictions and system verdicts that provides direct evidence of mental-model change rather than self-report alone.

2 Related Work

2.1 Student Reliance on AI and Verification Failure

The adoption of LLM chatbots in higher education has outpaced students' ability to critically evaluate outputs of these tools. Krupp et al. found that approximately half of ChatGPT solutions were mistakenly accepted as correct in a physics education context [10], while a survey of undergraduate healthcare students found that roughly 45% of students copied AI-generated text without verification [6]. Zhang et al. documented how students navigating plausible falsehoods in LLM outputs developed ad-hoc "sensemaking" practices. They typically re-query the same model rather than consulting external evidence [18]. Yeung et al. found that even students who understood LLMs can hallucinate and continued to use them in academic settings [17] without verifying the contents, demonstrating that knowledge of the problem does not translate into behavioral change.

These failure patterns are amplified by cognitive offloading. Kool et al. established that people systematically avoid effortful cognitive work when a low-effort alternative is available [8]. Fan et al. extended this to the AI context with the concept of *metacognitive laziness*: students increasingly delegate self-monitoring—the judgment of whether AI output is correct—to the AI system itself, effectively offloading verification [5]. Kosmyna et al. found neuroimaging evidence that AI-assisted essay writing reduced neural engagement and recall [9]. Taken together, these findings establish that the problem is not lack of knowledge but lack of scaffolded practice: students need an environment that makes verification the default behavior rather than an effortful detour.

2.2 Automated Verification and Hallucination Detection Tools

The HCI and NLP communities have developed a range of tools to surface LLM reliability signals.

HILL [12] is a hallucination identifier that assigns confidence scores to individual claims and surfaces links to external sources. A controlled study demonstrated that users could identify problematic claims more accurately when these signals were available—but HILL also surfaced a "false security" effect: participants who saw green confidence indicators suppressed their own evaluation, sometimes trusting incorrect claims because the system appeared to endorse them. This finding is directly relevant to VerifyChat's design: automation of the verdict without engaging the user's own reasoning can hold off their critical thinking when using these tools.

RELIC [2] takes a complementary approach, using self-consistency checking to highlight claims that the model itself is inconsistent about across multiple outputs. RELIC's study and evaluation showed improved error identification; its insight that keeping the chat response *fully readable* while verification signals appear in a separate panel informed VerifyChat's split-panel architecture. Neither HILL nor RELIC has been released as a publicly deployable artifact, limiting their practical accessibility.

Singh et al. explored embedding metacognitive prompts into AI-assisted search interfaces to nudge users toward more critical engagement [15]. Their work demonstrated that even lightweight scaffolding can shift how users interact with AI outputs, though without claim-level analysis, automated verdicts, or an integrated chat experience.

2.3 Commercial Verification Tools and Their Limitations

Several commercial products attempt to support verification in more accessible forms. Perplexity AI [14] provides inline citations for generated answers, allowing users to trace claims to external sources, but it does not decompose responses into discrete verifiable claims, does not assign reliability verdicts, and does not prompt active evaluation. Facticity AI [4] and Exa's [3] Hallucination Detector go further by flagging unsupported claims, but both require users to leave their current interface (via a browser extension popup or a copy-paste website), imposing the context-switching friction that inhibits adoption in a population already inclined toward least-effort paths [8].

2.4 Cognitive Forcing Functions and Metacognitive Scaffolding

The predict-then-reveal mechanism at the heart of VerifyChat draws on two converging lines of research. Buçinca et al. demonstrated that cognitive forcing functions—interface elements that require deliberation before the user can see an AI recommendation—significantly reduced overreliance in AI-assisted decision-making, particularly for users high in need for cognition [1]. Their work focused on structured decision tasks; VerifyChat extends this paradigm to the open-ended, conversational context of AI-generated academic content. Tankelevitch et al. documented the metacognitive demands that generative AI places on users and showed that prompting self-assessment improves critical evaluation of AI outputs [16].

Lee et al. found that AI tool use was associated with self-reported reductions in cognitive effort and critical thinking across a large sample of knowledge workers, underscoring the urgency of designing tools that preserve rather than erode these skills [11].

2.5 Research Gap

Considering the existing literature establishes five limitations that VerifyChat is designed to address: (1) automated tools risk false security by bypassing users' own reasoning; (2) no system combines automated verification with metacognitive scaffolding inside

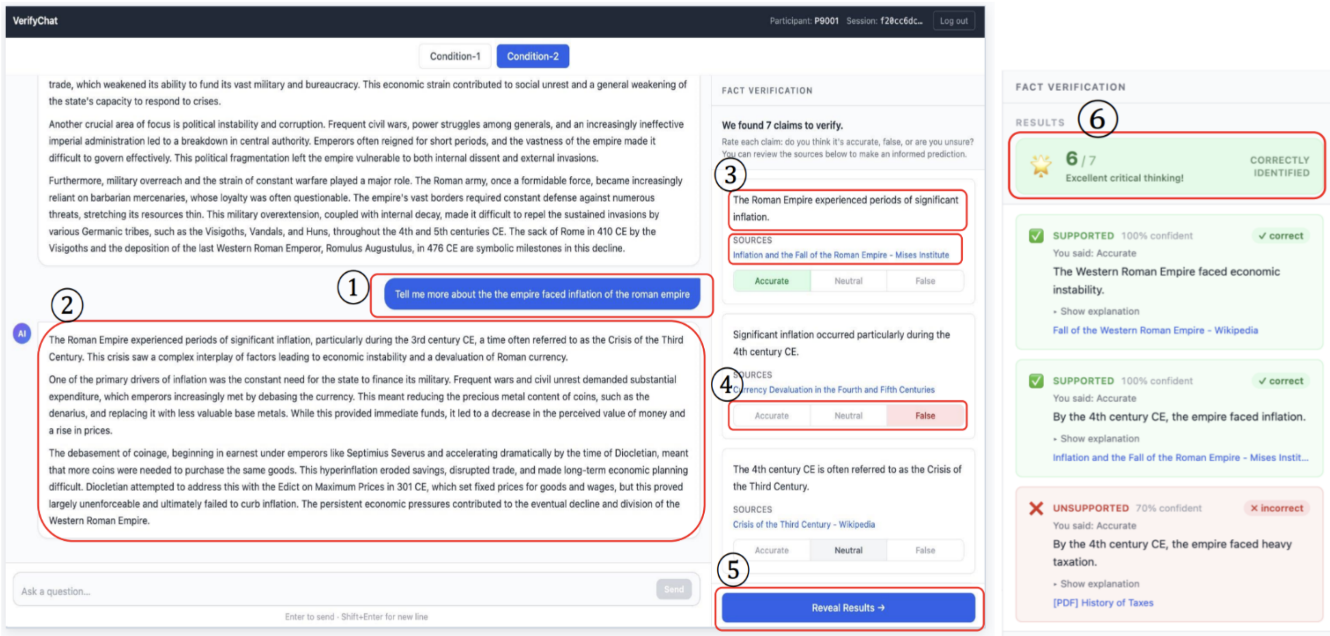

Figure 1: VerifyChat, a web-based claim-level verification system embedded directly in the chat workflow — scaffolds students to develop verification mental models for LLM-generated content. (1) The student poses a question to the AI in the chat panel. (2) The AI's response is streamed into the left panel exactly as in a standard chat interface. (3) In parallel, an automated pipeline decomposes the response into atomic checkworthy claims, which appear in the right-hand Fact Verification panel together with retrieved web sources for each claim. (4) Before seeing any system verdicts, the student commits a personal judgment on each claim — Accurate, Neutral, or False — implementing a cognitive forcing function that prevents passive over-reliance on automated checks. (5) The student clicks Reveal Results to expose the system's verdicts alongside their own predictions. (6) Per-interaction feedback shows how many claims the student correctly identified, along with claim-level outcomes (Supported / Unsupported / Insufficient Evidence), system confidence, and links to the supporting evidence, giving the student a behavioural signal of their developing verification skill.

the chat workflow; (3) existing research prototypes are not deployable; (4) commercial tools require context-switching friction that suppresses adoption; and (5) all existing studies use single-session designs, leaving longitudinal mental-model change entirely unmeasured. VerifyChat targets these limitations simultaneously through three interlocking contributions described in the following section.

3 System and Solution Design

3.1 Design Goals

VerifyChat was designed around three goals that emerge directly from the limitations identified in the related-work review:

(1) **Zero context-switching.** Verification must be available without leaving the chat, so the marginal cost of checking a claim approaches zero.
(2) **Require commitment before revelation.** Students must express their own judgment before seeing automated verdicts, preventing passive consumption [1].
(3) **Show evidence, not just labels.** Verdicts must be accompanied by retrievable source snippets so students can inspect the basis for each judgment, preventing the false-security effect observed in HILL [12].

3.2 Interface Architecture

VerifyChat is a single-page web application with two parallel regions:

- **Chat Panel (left).** Displays the conversational exchange in a clean, uncluttered format with no inline markers. The response remains fully readable while verification proceeds in parallel—a design principle drawn from RELIC's evaluation findings [2].
- **Verification Panel (right).** Structures the verification process through a two-step Predict-Then-Reveal cycle (Section 3.3) Fig 1.

3.3 Overall workflow of our system and the predict-then-reveal cycle

Every AI response triggers a six-step workflow (Fig: 1):

(1) Student submits a question through the chat interface.
(2) The AI response streams in and is displayed in the Chat Panel.
(3) Claims are automatically extracted from the response and presented as discrete, checkable items in the Verification Panel. Each claim is accompanied by the source URL that

will be used to verify it, reducing the need for external navigation.

(4) **PREDICT step**: Before any verdict is revealed, the student is prompted: *"The checker found k claims to verify. Before seeing results, which claims do YOU think might be inaccurate?"* The student selects checkboxes and optionally types brief reasoning.
(5) Student clicks "Reveal Results." Verdicts populate progressively (Supported ✓/ Unsupported × / Insufficient Evidence ~) alongside expandable evidence snippets with source links.
(6) The student receives immediate feedback: *"You correctly identified m of n flagged claims."*. The conversation continues and the cycle repeats for the next AI response.

This cognitive forcing function is directly grounded in Buçinca et al., who showed that requiring deliberation before seeing AI recommendations significantly reduced overreliance in decision-making [1], and in Tankelevitch et al.'s finding that prompting self-assessment at the claim level improves critical evaluation of generative AI outputs [16]. An end-to-end workflow of the system can be found in the following link : VerifyChat Demo

3.4 Technical Implementation

Verification Pipeline. For extracting checkable claims and extract the corresponding sources we followed a LOKI-inspired architecture [12] with five stages:

1. *Decomposer*: Gemini-Pro extracts atomic, checkworthy claims.
2. *Checkworthiness filter*: Gemini-Pro scores each claim for verifiability and retains the top 5–8 (preventing alarm fatigue).
3. *Query generator*: Gemini Pro generates search queries per claim.
4. *Evidence retriever*: Serper API is used to extract relevant sources using semantic similarity.
5. *Claim verifier* Gemini Pro assesses each claim against retrieved evidence and assigns a three-way verdict (Supported / Unsupported / Insufficient Evidence).

All the information and interaction with users and claims are stored in tables for sessions, messages, claims, predictions, verdicts, and user interaction events.

3.5 Key Design Decisions and Tradeoffs

The following table (Table: 1) includes all the design decisions we considered while designing the interface.

3.6 Scope and Limitations of the System

VerifyChat is designed for *factual, claim-rich* academic queries (e.g., history, science, current events). Its pipeline is less reliable for argumentative or value-laden content where "correctness" is inherently contested. Because of the financial constraints, we used Gemini-Pro as our claim decomposition and scoring, so the overall pipeline accuracy is bounded by Gemini's claim decomposition quality and by the coverage of web evidence retrieved via Serper, meaning the system can mis-flag true claims or miss false ones. To make the study meaningful, the AI was intentionally prompted to include injected hallucinations so that measurable verification opportunities were present; real-world use may differ.

Table 1: Key design decisions and their rationale.

Decision	Rationale & Grounding
Built-in chat	Eliminates copy-paste friction; suppresses adoption [8]
Predictions first	Cognitive forcing; requires System-2 engagement [1]
Show evidence	Avoids false security from labels [12]
Top 5–8 claims	Prevents alarm fatigue; [UX best practice]
3-way verdict	Avoids binary false confidence [7]
No inline markers	Keeps response readable; non-intrusive [2]

4 Methodology

4.1 Study Design

We conducted a within-subjects, single-session study. Every participant completed two consecutive phases in the same session: Phase 1 (standard chat baseline) followed by Phase 2 (VerifyChat). A within-subjects design was chosen to give every participant experience with both conditions, enabling paired comparisons with the small sample, and to observe any immediate strategy shift within the session. Moreover, for our study conducted the session online, which was participants' natural work environment.

4.2 Participants

We recruited undergraduate students ($N = 10$) through convenience and personal outreach at Stevens Institute of Technology. Among 10 participants, 8 participants were completed all the tasks completely without error. We anonymized the participants using P00x id. All the participants self-reported use of AI chatbots for academic tasks at least a few times per month. All participants were aged between 18 to 22 (mean = 20.13, SD = 1.25). Each session was around 30 minutes.

4.3 Procedure

Each study session followed a fixed sequence. After welcoming participants and obtaining written consent, we administered a pre-study survey collecting demographics, AI usage frequency, self-rated verification ability (1–7 Likert), and an open-text description of their current strategy for checking AI-generated content. Participants then completed two consecutive phases using the same Roman Empire history scenario. In Phase 1, participants used a plain AI chat interface with no verification panel. They were given three seed questions to start but were free to follow up however they wished. After each AI response, a free-text box prompted them to note any suspected inaccuracies before moving on; the facilitator observed but did not intervene. In Phase 2, participants switched to the VerifyChat interface and continued the same assignment. Here, each AI response triggered the Predict-Then-Reveal cycle (see section 3.3): participants read the extracted claims, selected

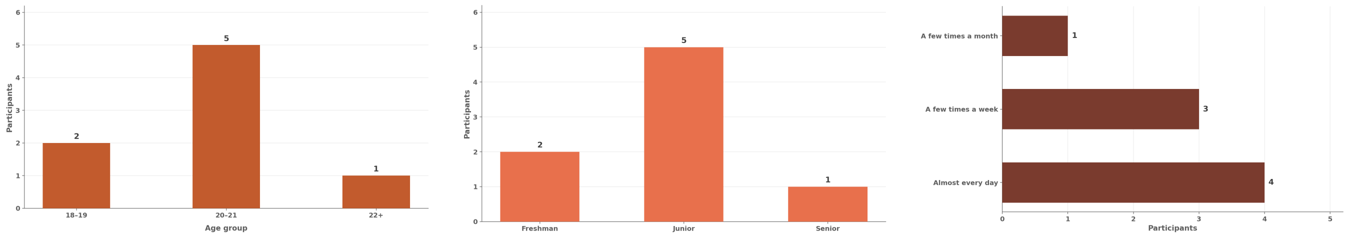

Figure 2: Different demographic and intro information of participants (Age group, Year of study and Prior AI experience)

those they believed might be inaccurate, clicked "Reveal Results," and inspected the automated verdicts alongside evidence snippets. The session closed with a post-study survey. We utilized Likert ratings on usability, helpfulness, and perceived behavior change, followed by a semi-structured exit interview probing how participants' verification strategies had shifted and how their trust in AI had changed across the two phases.

Scenario script. All participants received the same framing: *"Imagine you are a college student writing a short essay for your World History class about the Roman Empire. You will use two different AI chat interfaces one after the other. Please interact naturally, as you would at home working on a real assignment."*

To make the study easy and domain-constrained, we provided some seed questions that participants can use to start the task.

Seed questions. (1) What were the main reasons the Roman Republic transitioned into an empire under Augustus? (2) What major achievements in law, engineering, or culture did the Roman Empire leave behind? (3) What are the leading explanations historians give for the fall of the Western Roman Empire?

Injected hallucinations. As described earlier, to ensure measurable verification opportunities, the AI backend was prompted to include a fixed number of plausible but false claims. This guarantees ground-truth labels for F1 computation but means the hallucination rate is higher than typical real-world AI output (see 7 for more details).

4.4 Measurable metrics for our research questions

4.4.1 RQ1 — Verification Ease. Behavioral logs (claims inspected, evidence links opened), NASA-TLX four-subscale version (Mental Demand, Temporal Demand, Effort, Frustration), and five-item Likert usability scale administered after Phase 2 for usability of the system.

4.4.2 RQ2 — Mental Model Change. Pre-study open-text strategy description and semi-structured exit interview. Transcripts were analyzed with inductive thematic analysis.

4.4.3 RQ3 — Prediction Accuracy. For each AI response, we computed accuracy, and F1 score per participant for detecting the hallucinated claims in the fact-verification panel.

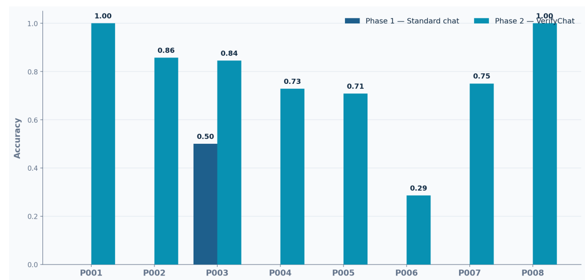

Figure 3: Hallucination detection accuracy of each participants in phase-1 and phase-2.

5 Data Analysis

We applied a mixed-methods approach for data analysis. First, we conducted a quantitative analysis on the changes in participants ability to detect hallucinated responses. We also ran statistical analysis on our result to check their significance. Moreover, We collected NASA-TLX Effort score from exit interviews to understand the cognitive load on users in both phases. During data analysis, we filtered the data where participants completed both task without any error.

5.1 Quantitative Findings

VerifyChat substantially improved participants' verification accuracy. To examine whether VerifyChat helped participants verify AI generated content more accurately, we compared per-participant mean accuracy between Phase 1 (Standard chat) and Phase 2 (VerifyChat). Accuracy was defined as the proportion of claims a participant judged correctly — either flagging an inaccurate claim as inaccurate or correctly accepting an accurate claim — averaged across each participant's interactions in a given condition. Across all 8 participants, mean accuracy rose from Phase 1 ($M = 0.026$, $SD = 0.18$) to Phase 2 ($M = 0.745$, $SD = 0.23$) (Fig. 4). The pattern was consistent across the sample: every participant exhibited higher accuracy in Phase 2 than in Phase 1. The aggregate per-interaction mean confirmed this pattern, rising from 0.026 across all Phase 1 interactions to 0.745 across all Phase 2 interactions (Fig. 4). Statistical tests for this comparison are reported in 5.2.

VerifyChat reduced participants' perceived cognitive load, most strongly on Effort and Frustration. To assess the cognitive cost of verification,

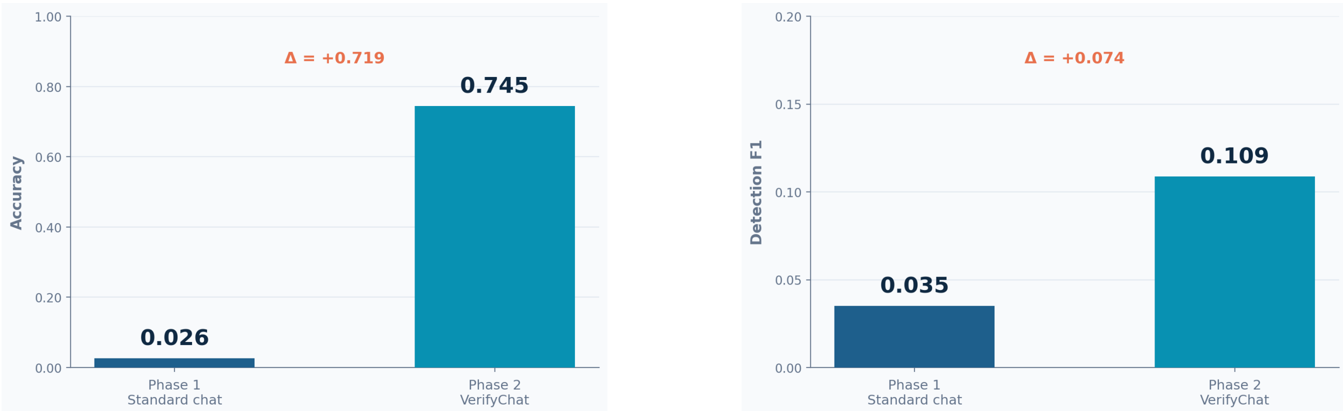

Figure 4: Hallucination detection performance across study phases. (Left) Mean accuracy in correctly judging claims rose from 0.026 in Phase 1 (Standard Chat) to 0.745 in Phase 2 (VerifyChat), an absolute improvement of Δ = +0.719. (Right) Mean detection F1 score improved from 0.035 to 0.109 (Δ = +0.074), reflecting gains in participants' ability to independently identify inaccurate claims before seeing automated verdicts

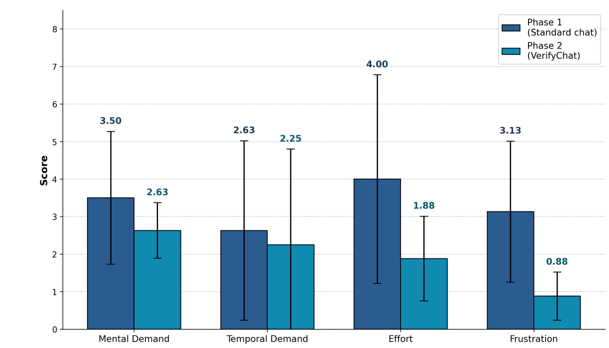

Figure 5: Mean workload scores (NASA-TLX) across four dimensions comparing the Standard chat baseline (Phase 1) and the VerifyChat interface (Phase 2).

we collected NASA-TLX ratings on four dimensions, Mental Demand, Temporal Demand, Effort, and Frustration after each phase, on a 0 to 10 scale. Overall mean ratings decreased from Phase 1 to Phase 2 on all four dimensions (Fig. 5). The largest reductions were observed for Frustration ($M = 3.13$, $SD = 1.88 \longrightarrow M = 0.88$, $SD = 0.64$; $\Delta = -2.25$) and Effort ($M = 4.00$, $SD = 2.78 \longrightarrow M = 1.88$, $SD = 1.13$; $\Delta = -2.13$). Mental Demand showed a smaller reduction ($M = 3.50$, $SD = 1.77 \longrightarrow M = 2.63$, $SD = 0.74$; $\Delta = -0.88$), while Temporal Demand was largely unchanged ($M = 2.63$, $SD = 2.39 \longrightarrow M = 2.25$, $SD = 2.55$; $\Delta = -0.38$). At the per-participant level, 6 of 8 participants reported lower Effort with VerifyChat than with the standard chat. Two participants (P005 and P006) reported higher Effort in Phase 2 than in Phase 1. We hypothesize that this pattern reflects differences in pre-existing verification strategy: both P005 and P006 entered the study without a source-checking habit, and may have experienced the per-claim verification step as additional work rather than as a substitute for a more effortful free-text strategy used in Phase 1.

Participants rated VerifyChat as easy to use and overwhelmingly preferred it over the standard chat. In the post-study survey, participants rated both interfaces on usability dimensions adapted from standard HCI usability scales (1 = strongly disagree, 5 = strongly agree). VerifyChat received a mean rating of 5.00 for ease of use, compared to 4.00 for the standard chat (Fig. 6). The verification panel itself was uniformly rated as easy to understand (M = 5.00), and participants strongly disagreed that the interface felt cluttered or overwhelming. When asked at the end of the session which interface they preferred, 7 of 8 participants selected VerifyChat. The one exception was P005, who entered the study without a source-checking habit, reported higher cognitive load with VerifyChat than with the standard chat, and described the predict-then-reveal step as "boring."

5.2 Statistical Analysis

From our analysis we found that mean accuracy improved from Phase 1 (Standard chat) (M = 0.026, SD = 0.18) to Phase 2 (VerifyChat) (M = 0.745, SD = 0.23). To ensure the robustness of these findings given the small sample size ($N = 8$), the improvement in accuracy between Phase 1 and Phase 2 , we evaluated both parametric and non-parametric methods. We conducted a paired t-test, a Wilcoxon signed-rank test, and a one-way repeated-measures ANOVA. We also calculated the effect size of the statistical significance using Cohen's d for paired samples. All tests confirmed a highly significant difference between the two phases: paired $t(7) = 7.47$, $p < .001$; Wilcoxon signed-rank $W = 0.0$, $p = .008$; and repeated-measures ANOVA $F(1, 7) = 55.81$, $p < .001$. And the effect size Cohen's d = 2.64. The results demonstrate a substantial and statistically significant improvement in accuracy from Phase 1 to Phase 2. The extremely large effect size ($d = 2.64$) suggests that the transition to the Phase 2 condition had a profound impact on performance. We like to mention that, given the limited sample size (N = 8), these results should be interpreted as descriptive findings from a pilot study rather than confirmatory evidence. A larger follow-up study with

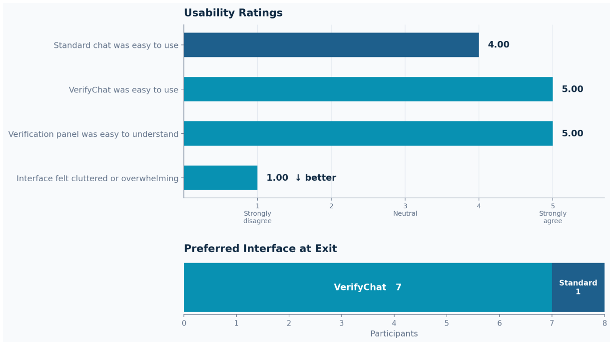

Figure 6: Post-study usability ratings and interface preference. (Top) Mean Likert scores (1 = Strongly Disagree, 5 = Strongly Agree) show VerifyChat rated higher on ease of use (M = 5.00) than the standard chat (M = 4.00), with the verification panel uniformly rated as easy to understand (M = 5.00) and the interface not perceived as cluttered or overwhelming (M = 1.00). (Bottom) Interface preference at exit: 7 of 8 participants selected VerifyChat as their preferred interface.

counterbalanced condition order and additional participants would be a better direction to confirm this directional pattern.

5.3 Qualitative Findings

Participants reported a shift toward source-driven, claim-level verification — though one participant disengaged from verification entirely. We analyzed exit interview transcripts and open-text responses to characterize how participants' verification practices changed across the two interfaces. Three patterns emerged consistently. First, participants reported a shift from holistic reading toward claim-by-claim evaluation. P003, a Computer Science junior, described the change explicitly: *"In the standard chat I mostly read the answer normally, but in VerifyChat I started looking at individual claims and thinking about which ones needed evidence."* P007, also in Computer Science, echoed this pattern: *"In the first interface I mostly read for whether the answer sounded correct, but in VerifyChat I started checking each claim more carefully."* Second, participants described source-checking as the strategy enabled by the verification panel. P002 noted that VerifyChat reduced verification to a single action: *"All I had to do was check the source to see if the information was reliable."* P008 described a similar effect: *"VerifyChat made me focus on whether individual claims were supported."* Third, the predict-then-reveal step was described as a slowing mechanism that promoted deliberation. P003 reported that *"it made me slow down and think before just accepting the system verdict,"* a description that mirrors the cognitive forcing function framing in the design. Counter to this pattern, P001, a Mechanical Engineering student, reported no change in process between the two interfaces and stated explicitly during the exit interview: *"I did not question the AI's accuracy."* This response coincides with the lowest behavioural engagement we observed — P001 marked all 19 claims as "Accurate" in Phase 2 — even though they rated VerifyChat 5/5 on usability and selected it as their preferred interface.

Across the eight participants, three distinct verification behaviours emerged. We conducted an inductive thematic analysis of the exit interviews, free-text strategy descriptions, and per-claim engagement logs. Following an open-coding pass, recurring themes were grouped into higher-level patterns describing how participants approached verification across the two interfaces. Three behavioural profiles emerged from this process, distinguished by (1) the participant's self-reported verification strategy prior to the study, (2) their qualitative description of how the two interfaces compared, and (3) their per-claim engagement pattern in Phase 2. Table 2 summarises the profiles and the participants assigned to each.

Table 2: Behavioural profiles that emerged from inductive thematic analysis of pilot participants' exit interviews, pre-study strategy descriptions, and Phase 2 engagement.

Profile	Description	Participants
Adopters	Entered with source-oriented strategies. Described a clear shift toward claim-level verification with VerifyChat. Engaged with the predict step and inspected sources.	P002, P003, P004, P007, P008
Resistant Verifiers	Entered without a source-checking habit. Found verification more effortful in Phase 2 than Phase 1 on NASA-TLX. Described the per-claim verification step as additional work rather than as a substitute strategy.	P005, P006
Passive User (False Security)	Liked VerifyChat and rated it highly, but did not engage with the verification step. Marked all claims as "Accurate" in Phase 2. Explicitly stated during exit interview that they did not question AI accuracy.	P001

6 Discussion

6.1 RQ1: Verification Ease

Usability and Cognitive Load. Participants rated VerifyChat as easier to use than the standard chat ($M = 5.00$ vs. $M = 4.00$ on a 5-point Likert scale) and uniformly agreed that the verification panel was easy to understand ($M = 5.00$). Mean NASA-TLX ratings decreased from Phase 1 to Phase 2 on all four dimensions, with the largest reductions on Frustration ($M = 3.13 \longrightarrow 0.88$) and Effort ($M = 4.00 \longrightarrow 1.88$).

Usability preference. Seven of eight participants stated a preference for VerifyChat over the standard chat at exit, including P001 who showed no behavioural change. The single dissenting participant (P005) reported higher cognitive load with VerifyChat than with the standard chat and described the predict-then-reveal step as "boring".

6.2 RQ2: Mental Model Change

We conducted an inductive thematic analysis of exit interviews, free-text strategy descriptions, and per-claim engagement logs. Three behavioural patterns emerged consistently across the eight participants.

Theme 1: Source-checking became the primary strategy. Multiple participants explicitly named source-checking as their new default verification approach after Phase 2. P002 says, *"All I had to do was check the source to see if the information was reliable."*

Theme 2: Shift to claim-level reasoning. Participants described moving away from holistic evaluation of the response toward examining individual claims. According to P003, *"In the standard chat I mostly read the answer normally, but in VerifyChat I started looking at individual claims and thinking about which ones needed evidence."*

Theme 3: Predict step forces deliberation. Participants reported that being required to commit a judgment before seeing the system's verdicts changed how they engaged with the AI response, echoing the cognitive forcing function framing in the design. According to P003, *"It made me slow down and think before just accepting the system verdict."*

Counter-pattern: Liked the tool, never engaged with verification. P001 reported no change in process between the two interfaces and stated explicitly during the exit interview: *"I did not question the AI's accuracy."* P001 marked all claims as "Accurate" in Phase 2 yet rated VerifyChat 5/5 on usability and selected it as their preferred interface. This pattern aligns with the "false security" effect documented in HILL [12].

6.3 RQ3: Prediction Accuracy

Hallucination-detection Accuracy. Mean accuracy improved from 2.6% in Phase 1 (standard chat) to 74.5% in Phase 2 (VerifyChat), an absolute improvement of approximately 71.9 percentage points. At the per-participant level, accuracy rose from a Phase 1 mean of $M = 0.026$ ($SD = 0.18$) to a Phase 2 mean of $M = 0.746$ ($SD = 0.23$), with every participant exhibiting higher accuracy in Phase 2 than in Phase 1. Mean detection F1 rose from 0.035 across all Phase 1 interactions to 0.109 across all Phase 2 interactions.

7 Limitations

We acknowledge several limitations in our current study design. First, our findings are exploratory due to a small, localized sample size of small participants. Because statistical tests are underpowered at this scale, generalization to broader populations is not warranted; a full study would require 20 to 30 participants to adequately measure the paired effect sizes observed here. Second, all measurements occurred within a single session lasting under one hour. Because genuine shifts in users' mental models are typically a longitudinal phenomenon, the strategy changes we observed may reflect in-session priming or demand characteristics rather than durable learning. Third, our automated verification pipeline is fallible. It relies on claim decomposition via Gemini Pro and web evidence gathered via Serper; if the system mis-flags a true claim, the student is inadvertently trained against a noisy oracle, introducing an uncontrolled covariate into our learning estimates. Finally, our study utilized a single domain—the Roman Empire—chosen specifically for its high fact-density and controlled knowledge space. Consequently, our findings may not seamlessly transfer to other contexts, such as STEM problem-solving, current events, or argumentative tasks. Furthermore, the artificially injected hallucination rate in this scenario was higher than typical real-world AI usage, which may have inflated the observed differences in performance between study phases.

8 Future Work

To address these limitations and expand upon our initial findings, future work must prioritize longitudinal and multi-domain evaluations. A multi-session study incorporating pre- and post-intervention drawing tasks would help the study to measure whether verified, claim-level feedback produces durable changes in how students perceive AI reliability, rather than mere in-session priming effects. Additionally, VerifyChat should be evaluated across diverse tasks—such as STEM homework, current events, and argumentative essays—to understand its efficacy in scenarios where correctness is contested or where relevant evidence is gated behind paywalls. Beyond study design, we also plan to explore iterative tool improvements to better support users. For example, to combat user patterns of blindly marking all claims as accurate without engagement, we could extend the design to adaptive scaffolding that could make the prediction step non-skippable or prompt explicit reflection after consecutive all-accurate sessions. Finally, we will investigate the impact of providing participants with source domain metadata (e.g., Wikipedia versus a specialized journal) prior to their predictions. This intervention could lower the difficulty of the prediction step for domain novices while preserving its deliberative educational value.

9 Conclusion

We presented VerifyChat, a chat-integrated fact-checking system designed to develop students' verification mental models through a predict-then-reveal interaction pattern. Rather than automating fact-checking in a way that encourages passive consumption, VerifyChat requires students to commit to a judgment about which claims are likely inaccurate before automated verdicts are revealed, grounding the interaction in the cognitive forcing function paradigm of Bucinca et al. and the metacognitive scaffolding principles of Tankelevitch et al. Our study with 10 undergraduate students found that hallucination-detection F1 and accuracy improved relative to a standard chat baseline, cognitive load dropped sharply, and almost all of the participants reported qualitative shifts toward claim-level and source-based reasoning strategies. Furthermore, one participant who rated the tool highly for usability while showing no behavioral change provided a direct replication of the false-security effect, demonstrating that positive UX and genuine metacognitive engagement are not equivalent. Combining all together, these findings support the potential of integrated, scaffolded verification as a path toward AI literacy that goes beyond tool dependency, while highlighting the need for longitudinal study designs and adaptive mechanisms to address users who resist engagement. VerifyChat's three contributions, an integrated verification interface, a predict-then-reveal cognitive forcing function applied to AI fact-checking, and a behavioral metric for mental-model change, provide a foundation for future work on how HCI can help students think more critically about the AI systems they increasingly rely on.

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A Survey and Interview Instruments

A.1 Pre-Study Survey Questions

Demographics.

(1) What is your age?
(2) What is your year of study? (Freshman / Sophomore / Junior / Senior / Graduate)
(3) What is your primary field of study?

Prior AI Experience.

(4) How often do you use AI chatbots for academic work? (Never / A few times a year / A few times a month / A few times a week / Almost every day)
(5) How would you rate your overall ability to judge when an AI response is reliable? (1–7)
(6) When you get an AI response for an academic task, what do you usually do to check if it is accurate? (Open text)

A.2 Post-Study Likert Survey (Phase 2)

Participants rated each item on a 5-point scale (1 = Strongly disagree, 5 = Strongly agree):

(1) VerifyChat was easy to use.
(2) The verification panel helped me identify potentially inaccurate claims.
(3) The predict step was useful.
(4) Seeing evidence sources helped me evaluate AI responses better.
(5) I would use a tool like VerifyChat for real academic tasks.
(6) After using VerifyChat, I feel more aware of how to check AI-generated information.

A.3 NASA-TLX Items (Abbreviated, 0–10 scale)

(1) **Mental Demand**: How mentally demanding was verifying the AI's response?
(2) **Temporal Demand**: How hurried or rushed was the pace of the task?
(3) **Effort**: How hard did you have to work to verify the AI's responses?
(4) **Frustration**: How frustrated, discouraged, or annoyed did you feel when using the tool?

A.4 Semi-Structured Exit Interview Questions

(1) When you were asked to predict which claims might be wrong, how did you decide?
(2) Was there anything that surprised you when the verification results were revealed?
(3) Did using VerifyChat change how you think about checking AI responses in general?
(4) Is there anything else you want to share about your experience?